



Data types and conversions

Previously, you learned about variables and how to name them. You learned that variables point to values that are stored in the computer's memory. In other words, the variables are like containers, and the values they store are their contents. Now, you're going to learn more about the values that your variables can contain. As you've learned, texts written between quotes in Python is called a string. Programs need to manipulate data, which can come in a lot of different forms or types. These data types include strings, integers, and floats.

First, a string is a sequence of characters and punctuation that contains textual information. Strings are instantiated with single or double quotation marks or the string function. This is an immutable data type, which means the values can never be altered or updated. An integer is a data type used to represent whole numbers without fractions. And float data types represent numbers that contain decimals. Most computers understand when you tell them to add two integers, or add two strings. But generally speaking, computers don't know how to work with different data types.

If you try to mix different data types, it can sometimes throw an error. As always, the computer tells us the cause of the error. It's like a little clue to help you improve your programming skills. Read the errors carefully, understand what they're trying to tell you,

and use that knowledge to fix the mistake. In this example, the last line of the error message says, "We've encountered something called a `TypeError`." Seven is being read as an integer and eight is being read as a string because of the quotation marks. No wonder we got an error: You can't add seven to a word.

As a data professional, you'll often need to aggregate a lot of data of different types. This will require converting the various types so they can be combined successfully. There's an effective method for doing this. But, first, it's important to know what you're working with. Python offers a helpful way to identify data types with the `type` function. You can use the `type` function to have the computer tell you the data type. For instance, the `type` function here tells us that `A` belongs to the `S-T-R` class, which is short for string. The number two belongs to the `I-N-T` class, or integer. And 2.5 belongs to the `float` class. As a reminder, a class is an object's data type. The class bundles data and functionality together. Now let's find out how to combine these different data types.

There are two ways to convert data in Python. First is implicit conversion.

Implicit conversion automatically converts one data type to another without user involvement. Here's an example. An arithmetic operations involving both integers and floats, the interpreter works in the background and converts integers to floats. You don't have to specify anything in your code to do this. However, if you want to convert numerical values to strings, you will need to do an explicit conversion. Explicit conversion is where users convert the data type of an object to a required data type. We use the predefined functions `I-N-T`, `float`, and `S-T-R`.

This is sometimes called typecasting, because the user casts or changes the data type. Let's use the `S-T-R` function inside the string we want to have interpreted as output. And

now the output of this calculation will be stored as a string. Great work. You'll continue exploring data types and seeking solutions to errors throughout this program.

Debugging, or figuring out what's wrong when your code won't work, is a really useful skill for any data professional. Also, as a final tip: All of us in the profession, even experienced data professionals and code developers, search for answers online when we encounter an error.

This is a common strategy and a huge time saver. Always look to the data community for answers and inspiration.

String

A sequence of characters and punctuation that contains textual information

Immutable data type

A data type in which the values can never be altered or updated

Integer

A data type used to represent whole numbers without fractions

Float

A data type that represent numbers that contain decimals

Question



Fill in the blank: The _____ data type is a sequence of characters and punctuation that contains textual information.

- ☒ string
- ☐ float
- ☐ integer
- ☐ assignment

✓ Correct

The string data type is a sequence of characters and punctuation that contains textual information.

Continue

Skip

You can use the `type()` function to have the computer tell you the data type.

Implicit conversion

Python automatically converts one data type to another without user involvement

Explicit conversion

Users convert the data type of an object to a required data type